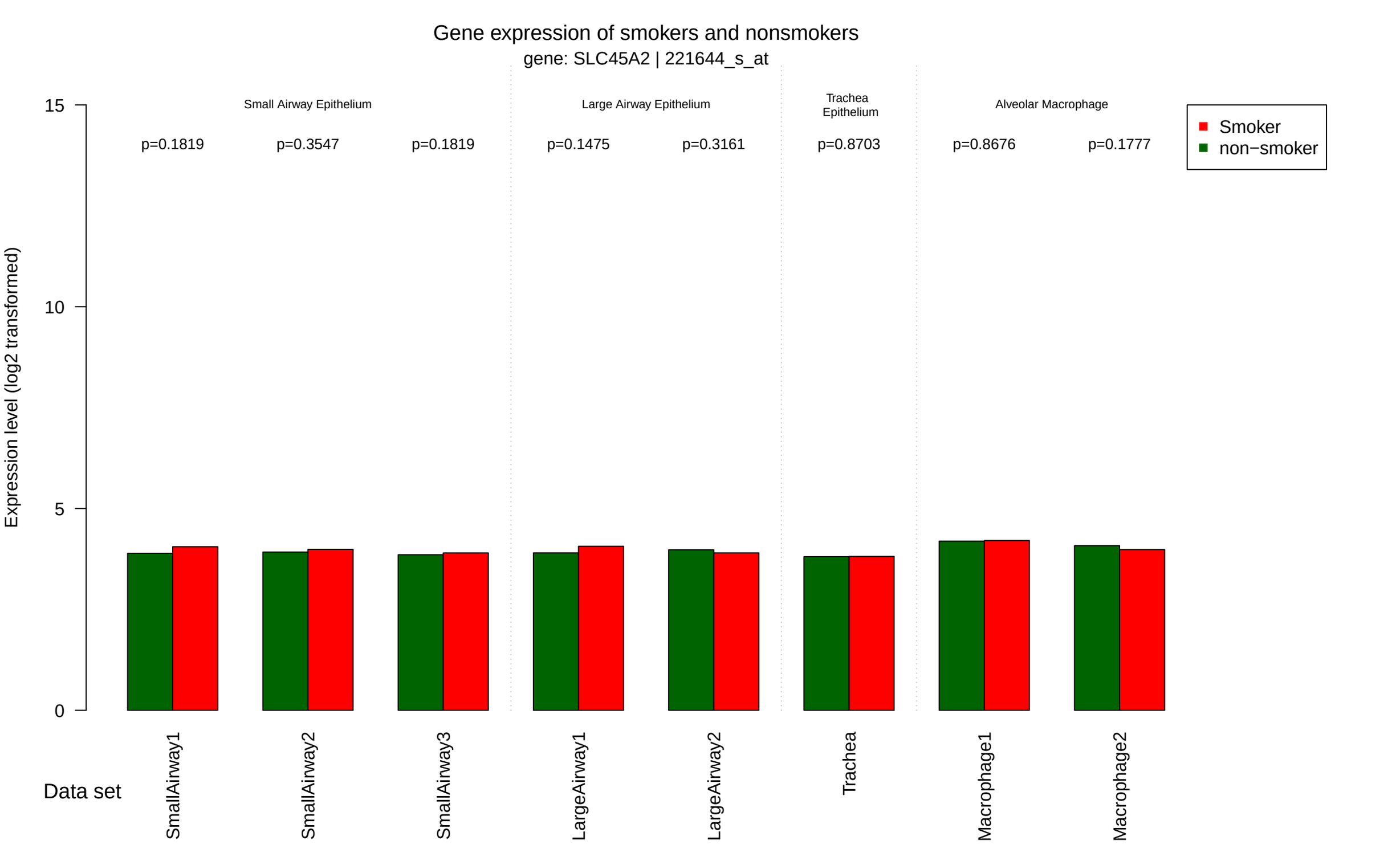
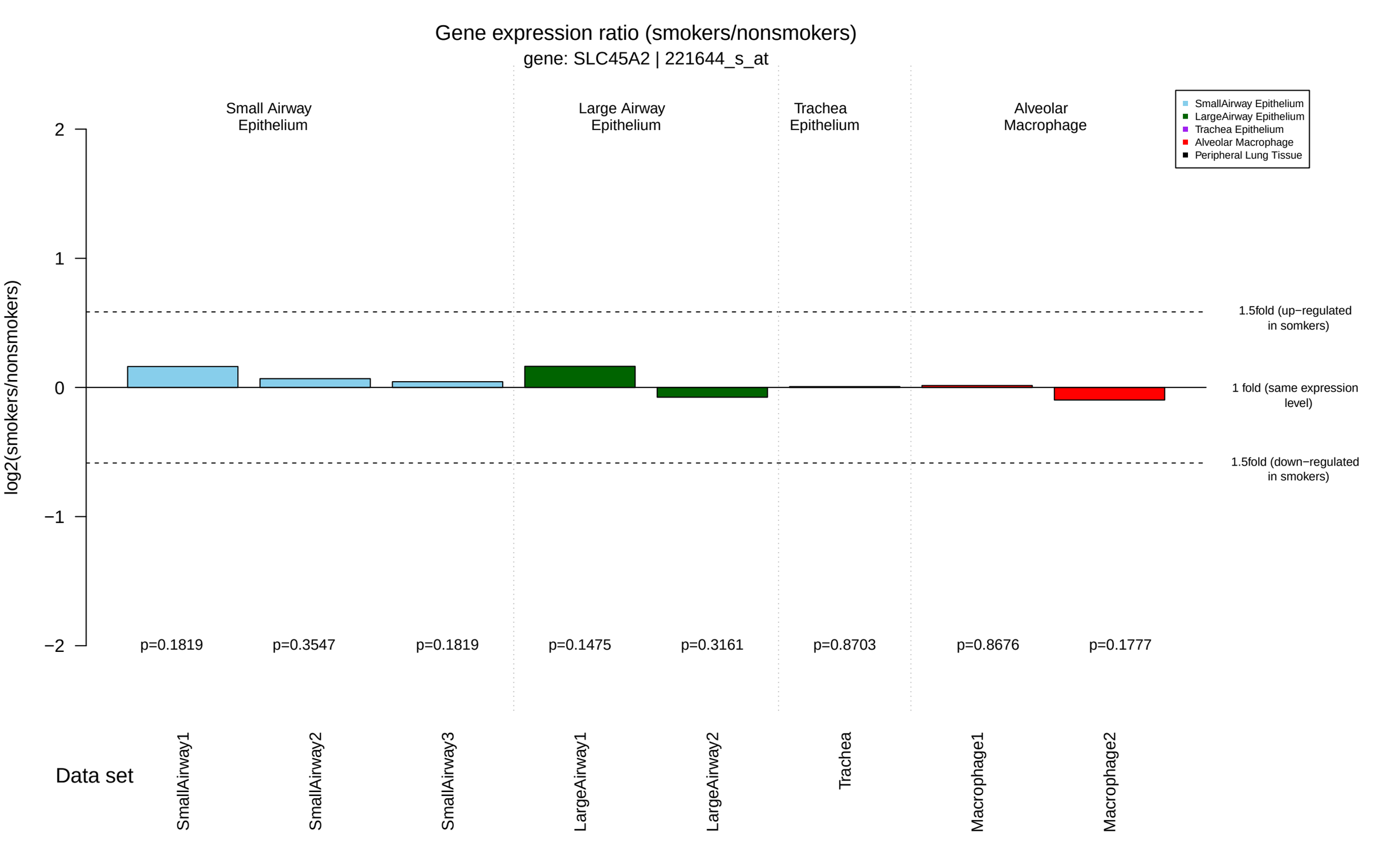
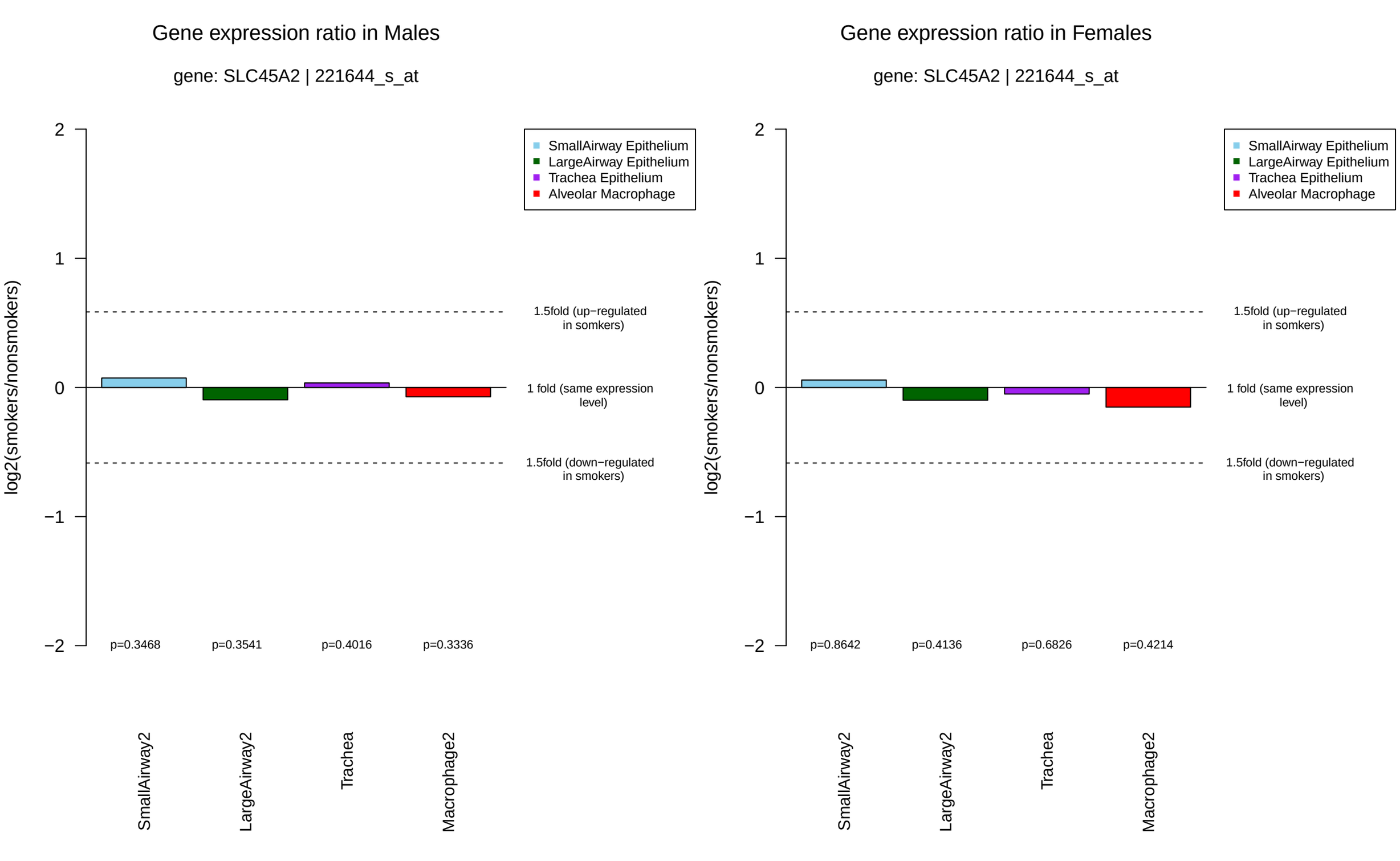
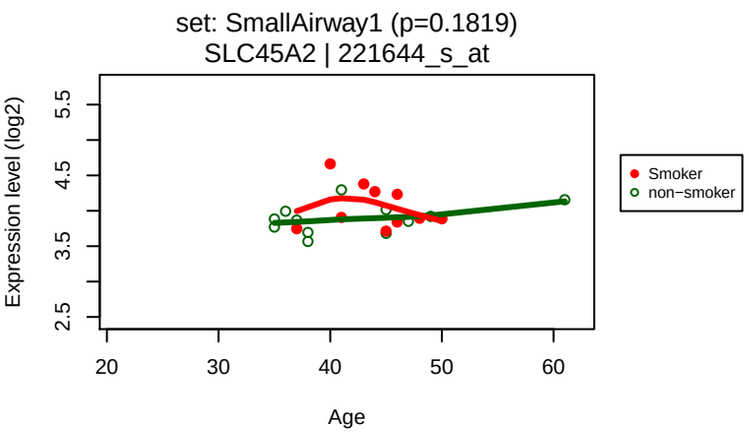
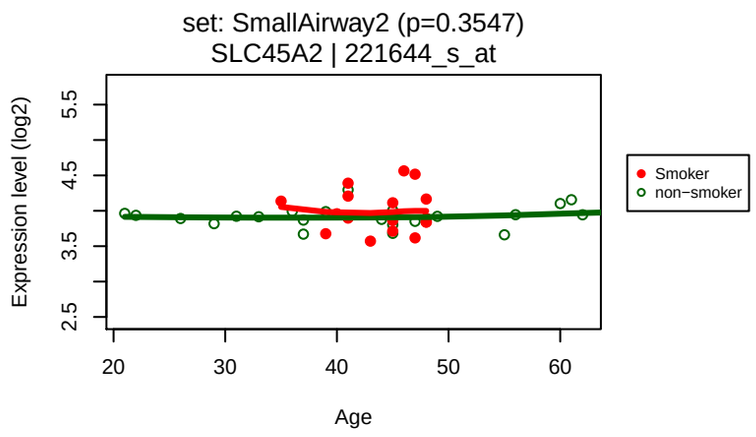
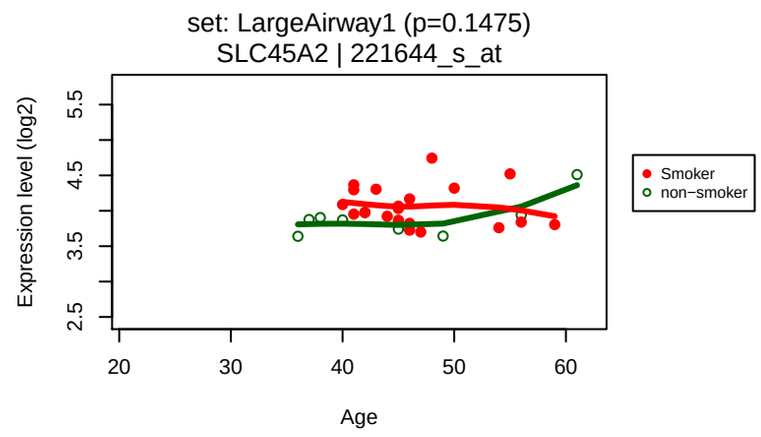

Gene expression of smokers and nonsmokers
gene: SLC45A2 | 221644_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1819
p=0.3547
p=0.1819
p=0.1475
p=0.3161
p=0.8703
p=0.8676
p=0.1777
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

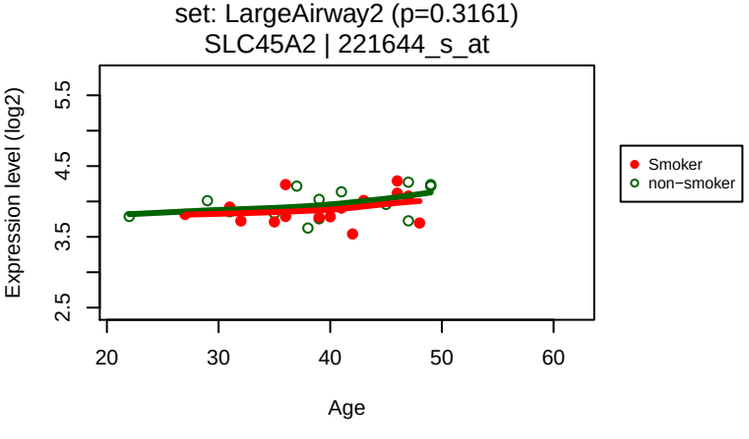

Gene expression ratio (smokers/nonsmokers)
gene: SLC45A2 | 221644_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1819
p=0.3547
p=0.1819
p=0.1475
p=0.3161
p=0.8703
p=0.8676
p=0.1777
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

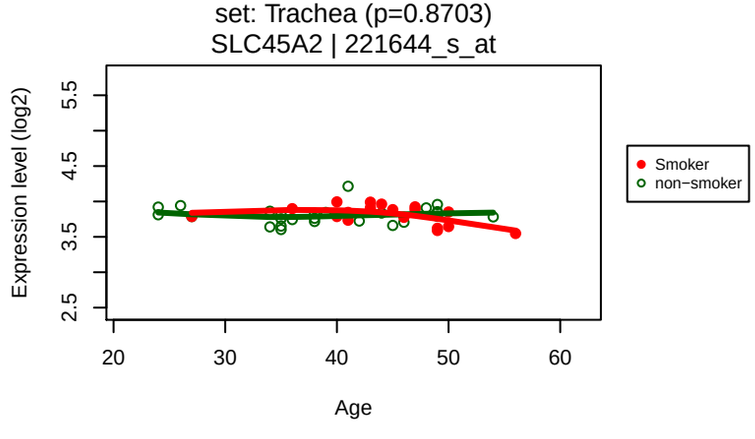

Gene expression ratio in Males
gene: SLC45A2 | 221644_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3468
p=0.3541
p=0.4016
p=0.3336
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC45A2 | 221644_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8642
p=0.4136
p=0.6826
p=0.4214
SmallAirway2
LargeAirway2
Trachea
Macrophage2

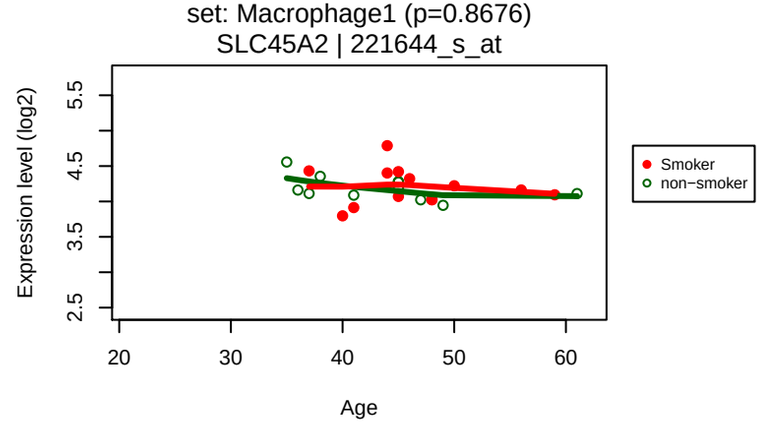

set: SmallAirway1 (p=0.1819)
SLC45A2 | 221644_s_at
Expression level (log2)
Age
Smoker
non−smoker

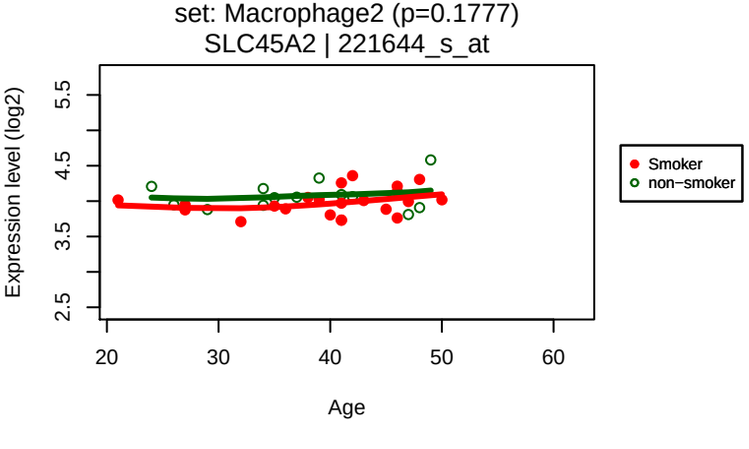

set: SmallAirway2 (p=0.3547)
SLC45A2 | 221644_s_at
Expression level (log2)
Age
Smoker
non−smoker

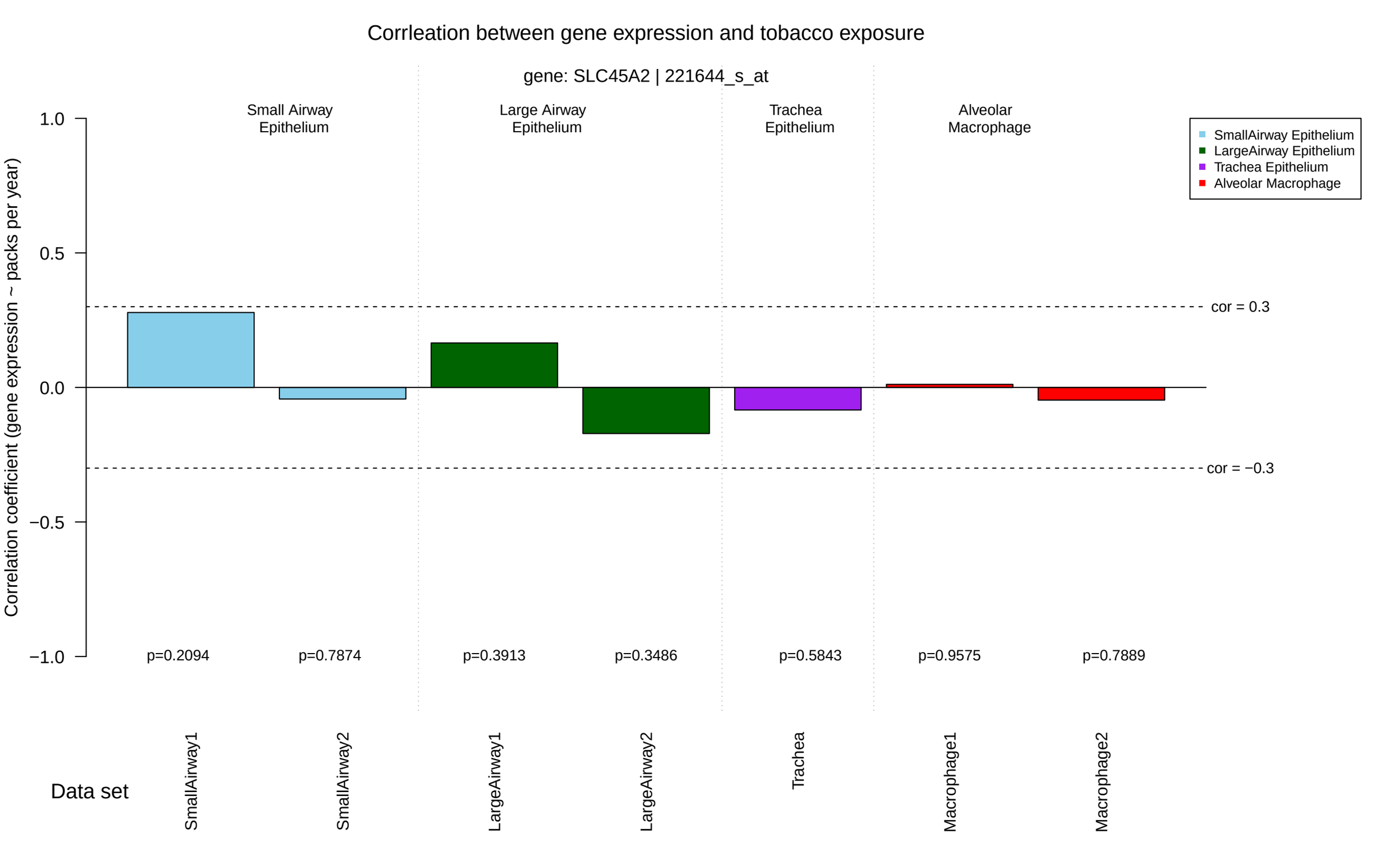

set: LargeAirway1 (p=0.1475)
SLC45A2 | 221644_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3161)
SLC45A2 | 221644_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8703)
SLC45A2 | 221644_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8676)
SLC45A2 | 221644_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1777)
SLC45A2 | 221644_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC45A2 | 221644_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2094
p=0.7874
p=0.3913
p=0.3486
p=0.5843
p=0.9575
p=0.7889
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2